INCH-POUND
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28 April 1989
SUPERSEDING
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MILITARY SPECIFICATION

PROPELLANT, STICK, M31A1E1 FOR USE IN CHARGE PROPELLING M203A1

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M31A1E1 stick propellant for use in the 155MM propelling charge, M203A1 (zone 8) for the 155MM Howitzer, M198.

1.2 Classification. The propellant shall be Type II, cylindrical, slotted, single-perforated stick.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extentspecified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-P-193 - Potassium Sulfate (For Ordnance Use)
MIL-D-218 - Dibutylphthalate (For Use In Explosives)
MIL-N-244 - Nitrocellulose
MIL-N-246 - Nitroglycerin

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-255 - Ethyl centralite (carbamite)
MIL-C-306 - Carbon black, powder, dry
MIL-N-494 - Nitroguanidine (Picrite)
MIL-Q-9858 - Quality Program Requirements
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-70462 - Charge Propelling, 155MM, M203A1 Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection and Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9345104 - Propellant and Liner Assembly
9390418 - Packing and Marking for Box, Wood, Nailed for Box, Fiberboard, for Propellant, Stick, for Charge, Propelling, 155MM M203A1
9390472 - Barrier bag
9390474 - Packing and Marking for Box, Fiberboard for Propellant Stick M31A1E1

(Copies of specifications, standards, drawings, publications and other Government documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take

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precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Constituent material. The constituent materials of the stick propellant shall comply with the requirements of the applicable specification as follows:

Constituent Material	Conform to Specification
Dibutylphthalate	MIL-D-218
Nitrocellulose	MIL-N-244, Type I, Grade A
Nitroglycerin	MIL-N-246, Type I
Nitroguanidine	MIL-N-494, Class I, Type II
Potassium Sulfate	MIL-P-193, Type I
Ethyl centralite	MIL-E-255
Carbon Black	MIL-C-306

3.2 Propellant compositions and chemical properties. The compositions and chemical properties of M31A1E1 stick propellant shall conform to the requirements of Table I when tested in accordance with the applicable test methods in 4.5.1.4.

Table I

Propellant compositions and chemical properties*

Properties	Percent
Nitrocellulose Type I, Grade A Cotton Linters only	21.50 ± 1.30
Nitroglycerin	18.00 ± 1.00
Nitroguanidine	54.70 ± 1.00
Potassium Sulfate	1.25 ± 0.25
Dibutylphthalate	3.0 ± 0.30
Ethyl Centralite	1.50 ± 0.10
Carbon Black Grade A	0.05 + 0.05
Total Volatiles	0.40 Maximum

*The composition shall be calculated on total volatiles free basis.

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3.3 Form and dimension. The form and dimension of the stick propellant shall conform to the requirements below when tested in accordance with test method in 4.5.1.2.

3.3.1. Form. The stick propellant shall be cylindrical, slotted, straight and single-perforated.

3.3.2 Dimension. The following requirements regarding stick propellant dimensions shall govern unless authorization for departure is given by the procuring activity concerned prior to manufacture of the propellant.

3.3.2.1 Length. The stick propellant shall be with length specified 28 7/8 inch to 29 1/8 inch.

3.3.2.2 Diameter. The diameter of stick propellant shall comply with either the mean variation or the standard deviation uniformity requirements shown in Table II.

TABLE II

Mean variation and standard deviation of individual dimensions expressed as a percentage of the mean dimension.

Dimension	Percent Mean Variation Maximum	Percent Standard Deviation Maximum
Diameter	3.125	4.75

3.3.2.3 Perforation diameter ratio. The average stick diameter (D) shall be approximately three times the average diameter of the perforation (d).

3.3.2.4 Web measurement. The standard deviation of the web measurements, expressed as a percent of the average web measurement, shall not be greater than 10 percent.

3.4 Weights

3.4.1 Average stick weight and weight standard deviation. The average stick weight and weight standard deviation of the propellant shall be determined in accordance with 4.5.1.1.

3.4.2 Stick propellant bundle weight. The weight of a bundle of one hundred (100) stick propellants shall be determined in accordance with 4.4.2.2.

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3.4.3 Charge weight loadability. The charge weight required to meet the ballistic performance as specified in paragraph 3.9 shall be capable of being loaded in the combustible cartridge case with an additive liner with propellant weight not more than 29.0 lb. In addition, the contractor responsible for manufacturing of stick propellant shall perform the trial loading in accordance with 4.5.1.5 and the 29-lb stick propellant bundle with an additive liner shall be loaded into the combustible cartridge case or an inert cartridge case without any difficulty.

3.5 Stability. For heat test conducted at 120 degrees centigrade, the propellant shall not change the color of the methyl violet test paper to a salmon pink color in less than 40 minutes when tested in accordance with 4.5.1.4.

3.6 Heat of explosion. Heat of explosion of the stick propellant shall be 850.0 ± 30.0 cal/gm when determined in accordance with 4.5.1.4.

3.7 Density. The density of stick shall be 1.63 gm/cc minimum when determined in accordance with 4.5.1.4.

3.8 Closed bomb. Relative quickness and relative force at 90°F, 145°F and -40°F shall be determined in accordance with 4.5.2.

3.9 Proving ground

3.9.1 Proving ground assessments. The propellant when fired in a M198 Howitzer with a M199 cannon with 80% (minimum) remaining life and with the Projectile PXR-6325 (preferred to M549 inert or M549A1 if HE projectile is to be fired) with a 96.0 ± 0.1 pounds (lbs) as fired projectile weight, and charges manufactured in accordance with MIL-C-70462 (see 6.9) shall comply with the following requirements when tested as specified in 4.5.3. To meet requirements of lot formation of MIL-A-48078, the calibration lot charges will be loaded at the same time by the contractor responsible for charge manufacture.

3.9.1.1 Permissible individual maximum pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 57,950 psi at any operating temperature (-65°F to +145°F) when fired in a standard weapon (see 6.7).

3.9.1.2 145°F pressure. At standard weapon conditions and recommended charge weight, the average pressure plus 2.2 times the standard deviation shall not exceed 57,950 pounds per square inch (psi) at 145°F.

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3.9.1.3 Charge establishment phase. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of assessment to a velocity of 2710 feet per second (fps).

3.9.1.4 Uniformity phase

3.9.1.4.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 7.0 fps.

3.10 Pressure/temperature coefficient. The pressure/temperature coefficients between 70°F and 145°F and between 70°F and -65°F shall be determined in accordance with 4.5.3.4.

3.11 Blending process control. A propellant lot shall be a minimum of 75,000 lbs. The contractor shall assure that each lot has been sufficiently blended to meet the ballistic uniformity requirements (see 3.9). The contractor shall submit a blending process control document to the government specifying the blending process control steps which are considered critical for meeting the ballistic uniformity requirements. The blending process control document shall be submitted in accordance with 4.1.2.

3.12 Contract data requirements (see 6.2)

3.13 Preproduction sample inspection and first article test (see 4.3). This specification contains technical provisions for preproduction sample inspection and first article test. Requirements for submission of preproduction samples and first article test samples by the contractor shall be as specified in the contract.

3.14 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.1.2 Submission of blending process control document. The contractor shall prepare a process control document which details all of the specific procedures involved in propellant blending processing operations. As a minimum, the document shall detail the procedures of accumulation of sublots prior to initiating final blending. This document shall be part of the contractor's quality program requirement in accordance with MIL-Q-9858. It shall be submitted to the contracting officer for review by the technical agency. The contractor shall not initiate production without written approval of the blending procedures from the contracting officer. Once approved the contractor shall not change the process without written approval from the contracting officer (6.13).

4.2 Classification of inspections. The following inspections shall be conducted on this material:

a. Preproduction sample inspection and first article test (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Preproduction sample inspection and first article test.

4.3.1 Preproduction sample (pilot lots). The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1.1 Submission. The contractor shall select propellant granulations for approval by Government Engineering and Product Assurance, which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit a

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preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.1.2, usually consisting of up to three pilot lots of approximately 2500 lbs each, representing each of the candidate granulations.

4.3.1.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1.1 may be subjected by the Government to any or all of the requirements of this specification.

4.3.1.3 Rejection. Government Engineering and Product Assurance personnel and the contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contracting Officer shall designate the preproduction sample as acceptable and shall authorize the Contractor to produce the selected granulation for regular production. If the pilot lots submitted fail to meet or bracket the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1.1 Until the preproduction sample is designated acceptable, the Contractor is not authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer. All future production lots should be of the same basic granulation as the selected granulation authorized for regular production.

4.3.2 First article test. The first article test is a one time qualification test for a new contractor to produce acceptable stick propellants. Once a contractor has demonstrated the ability to produce acceptable stick propellant the first article test may be waived by the Contracting Officer.

4.3.2.1 Submission. Test samples shall be withdrawn from the first lot produced by the contractor and tested in accordance with all the requirements of this specification and with 4.3.2.2. Until the first article test has been performed and found to be acceptable, the contractor is in no way authorized by the Government to initiate second lot production unless otherwise directed by the Contracting Officer.

4.3.2.2 Testing to be performed. The following test program shall be used for the first article test.

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Table III
First article test

Test Phase	Temp (oF)	No of Controls Rds	No of Test Rds	TECOM TOP
Propellant Checkout	70		10	TOP-4-2-504
	145		10	TOP-4-2-504
	-60		10	TOP-4-2-504
Sequential Environmental	145	15	60	TOP-4-2-504
	-60	15	60	TOP-4-2-504

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.4.2 Examination and testing

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing. Ten (10) containers shall be selected at random from each lot of propellant. 1.8 lbs of propellant shall be removed from each container and mixed to form a composite sample of eighteen (18) pounds. Eighteen (18) pounds of the sample is intended to be used as follows:

4 lb for physical and chemical properties tests in accordance with 4.5.1.

9 lb for closed bomb in accordance with 4.5.2.

5 lb shall be used for 65.5 °C surveillance test in accordance with Method 407.1, MIL-STD-286.

All samples shall be packed in air tight containers and shall be marked to show the propellant designation, lot number, manufacturer, date of sampling, and number of pounds in the lot. If the sample fails to comply with the requirements, the lot shall be rejected.

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4.4.2.2 Sampling for stick propellant bundle weight (see 6.2). Ten (10) propellant boxes shall be selected at random from each lot of propellant. One hundred (100) propellant sticks shall be removed from each box to form a bundle. The bundle shall be weighed to the nearest gram. Ten (10) individual bundle weights from the ten (10) selected boxes shall be reported in the Propellant Description Sheet.

4.4.2.3 Sampling for packaging inspection. Sampling for packaging inspection shall be done in accordance with MIL-STD-105D. MIL-STD-1235 may be used if approved by procuring activity.

4.4.2.3.1 Barrier bag (see Dwg 9390472).

Categories	Defects	AQL	Method of Inspection
Critical	None		
Major 101	Inside bag dimension not proper (length and width)	0.40%	Gage
102	Seam strength not proper	Note 1	4.4.4.2 of MIL-P-116
Minor 201	Workmanship	0.65%	Visual

Note 1. Two (2) heat-sealed specimens per machine, per shift and one (1) heat-sealed specimen after the machine is turned on or whenever there is a change in operating temperature or cycle time of the machine.

4.4.2.3.2 Packing and marking for box, fiberboard for propellant stick M31A1E1 (see Dwg 9390474).

Categories	Defect	Sample Size	Method of Inspection
Critical	None		
Major 101	Seam strength not proper	See Note 1	4.4.4.2 of MIL-P-116

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Minor			
201	Box corners not blunted	See Note 2	Visual
202	Air not evacuated from sealed bag (80% minimum of bag not against box)	See Note 2	Visual
203	Workmanship	See Note 2	Visual

Note 1. Three (3) samples shall be withdrawn per machine, per shift (one at the beginning, one at the middle and one at the end) and one (1) sample after the machine is turned on or whenever there is a change in operating temperature or cycle time of the machine. If any sample is found to be unacceptable, corrective action must be taken and another sample shall be tested and found to be acceptable prior to continuation of the packing operation. In addition, if any sample shows areas where there is no evidence of bonding, previous production will be screened starting with the last bag sealed immediately prior to the failed sample and working backwards through production. Screening will continue until an acceptable sample has been found. An acceptable sample during screening is one that shows evidence of bonding at all areas.

Note 2. Three (3) samples shall be withdrawn per shift (one at the beginning, one at the middle and one at the end). If any sample is found to be unacceptable, corrective action must be taken and another sample shall be inspected and found to be acceptable prior to continuation of the packing operation.

4.4.2.4 Sampling for ballistic testing. Appropriate amount of containers shall be selected randomly from each lot of propellant for ballistic testing at each temperature as specified in 4.5.3. The total sample size at each temperature shall consist of the weight in pounds as specified on the applicable assembly drawing multiplied by the sample size times the factor 1.3. The sample shall be selected from individual containers.

4.4.2.4.1 Aging of stick propellants. The stick propellants shall be aged a minimum of forty-five (45) days at sealed and ambient conditions from the date of the packing (unless other conditions specified), prior to ballistic teting.

4.4.2.5 Permissible individual maximum pressure. If at any time during testing, an individual pressure greater than that specified in 3.9.1.1 is observed, testing shall cease and the lot shall be rejected. If, at any temperature firings, the average pressure plus three standard deviations exceeds the individual pressure requirement, the lot shall be referred.

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4.4.2.5.1 145°F pressure. If the average pressure plus 2.2 times the pressure standard deviation at 145°F exceeds 57,950 pounds per square inch (psi), the lot shall be rejected.

4.4.2.6 Charge establishment (see 3.9.1.3). If the lot does not meet the applicable requirement, testing shall cease and the lot shall be rejected.

4.4.2.7 Velocity uniformity (70°F) (see 3.9.1.4.1 and 6.4). If the product of the factor (0.72) times the velocity standard deviation of the rounds tested in accordance with 4.5.3.2 exceeds the applicable requirement, the lot shall be rejected.

4.4.3 Charge weight loadability. Loadability characteristics shall be verified by the ballistic samples as specified in paragraph 4.4.2.4 and tested in accordance with 4.5.3. If the charge weight required to meet the ballistic performance (see 3.9) can not be loaded in the combustible cartridge case with an additive liner or the recommended charge weight is greater than 29 lbs, the lot shall be rejected.

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed.

4.5 Methods of inspection.

4.5.1 Physical and chemical properties testing (see 4.4.2.1)

4.5.1.1 Average stick weight and weight standard deviation (see 6.2). Thirty (30) sticks shall be selected at random from the 4 - lb sample and each stick shall be weighed to the nearest gram. The average stick weight and weight standard deviation shall be calculated as follows:

$$\text{Average stick weight} = \frac{\Sigma Xi}{30}$$

Xi = weight of each individual stick

$$\text{Standard deviation (s)} = \sqrt{\frac{n \sum x^2_i - \left(\sum x_i \right)^2}{n\,(n-1)}}$$

n = total number of sticks

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4.5.1.2 Dimensions (see 6.2). The 30 sticks after weighing (see 4.5.1.1) shall be used again and tested for dimensions in accordance with 504.1, MIL-STD-286 with the exception that the length shall be inspected in accordance with 4.5.1.3. Failure to meet any applicable requirements shall be cause for rejection of the lot.

4.5.1.3 Length. Five hundred (500) sticks shall be randomly selected from each sublot and measured by gage for length requirement. If one (1) stick is found to be exceeding the maximum length requirement or three (3) sticks are found to be below the minimum requirement, the sublot shall be rejected. A 100% inspection on stick length shall be done on the rejected sublot. Each stick shall be measured by gage and any stick which does not meet the length requirement should be removed from the sublot. The final blending on the entire lot should not be initiated until all sublots are found to be meeting the stick length requirement. The gage design for stick propellant length measurement shall be submitted to the technical agency for approval.

4.5.1.4 Chemical and physical testing. The remaining 4 - lb sample after withdrawal of 30 sticks shall be divided into 1 - inch pellets (approximate). The pellets shall then be thoroughly mixed to form a composite. The required specimen shall be randomly selected from the composite for the chemical and physical testings as specified in Table IV. Failure of any samples to meet the applicable requirements shall be cause for rejection of the lot.

Table IV

Methods from MIL-STD-286 for chemical and physical properties of the propellant.

Properties	Methods	Alternate
Nitrocellulose	209.2	
Nitroglycerin	208.3	208.1, 213.5
Nitroguanidine	213.5	213.1
Potassium Sulfate	316.1	
Dibutylphthalate	208.3	222.1, 208.4, 226.2

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Ethyl Centralite	202.2	
Carbon Black	See 4.5.1.4.1	
Total Volatiles	103.5	
Stability	404.1	
Heat of Explosion	802.1	
Density	T510.3	510.11

4.5.1.4.1 Carbon black. The actual percentage of carbon black shall be determined in accordance with Method 308.1, MIL-STD-286.

4.5.1.5 Charge weight loadability. Five (5) 29-lb stick propellant samples shall be randomly withdrawn from the final blended lot. Each 29-lb stick propellant sample shall be weighed to 29.0 lb - 2.0 ounces. Each 29-lb stick propellant sample shall be bundled , tied, and loaded into a rigid cyclinder that has an inside diameter of 5.75 - .10 inch and a length of 30.0 $\pm$ 1.0 inch. If any stick bundle cannot be loaded into the cylinder, the technical agency shall be notified. The disposition of the lot shall be withheld until the recommended charge weight (RCW) is determined by the proving ground. If the propellant lot at RCW can be loaded without any difficulty, the lot shall be accepted, provided it also meets all other requirements as specified in this specification. If the propellant lot at the RCW cannot be easily loaded, the lot shall be rejected.

4.5.2 Closed bomb test. From the 9 lb sample, 3 lb each shall be used for 90°F, 145°F and -40°F closed bomb test. All the test specimens shall be conditioned at the required temperature for a minimum of one hour with the following tolerances: 90°F from 88°F to 92°F; 145°F from 140°F to 145°F; -40°F from -40°F to -35°F. Tests shall be performed in the 700 c.c. closed bomb with stick propellant length of approximately 7 inches in accordance with the latest revision of Method 801.1 MIL-STD-286 and fired with reference stick propellant (see 6.12) for informational purpose.

4.5.3 Proving ground assessments (see 6.5). These tests shall be conducted at a Government proving ground in accordance

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maximum chamber pressure of each charge for information and ignition delay of each charge. Velocity coils will be used to measure the muzzle velocity of each charge. Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65°F, 145°F and 70°F conditioning: -65°F from -65°F to -60°F; 145°F from 140°F to 145°F; 70°F from 67.5°F to 72.5°F. Temperature indicator (thermometer) strips shall be placed on the side (combustible cartridge case) of all charges conditioned at 145°F as a further assurance of the proper conditioning temperature. If any indicator strip shows the charge has been exposed to a temperature greater than 145°F, that charge will not be fired. Propelling charges used for test as calibration shall be assembled in accordance with MIL-C-70462 (see 6.9).

4.5.3.1 Charge establishment

4.5.3.1.1 Charge slope method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) plus three calibration rounds shall be conditioned and fired at 70°F. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.9.1.3). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.3.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

4.5.3.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity curve can be established from test results generated. This curve can then be used as in 4.5.3.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into charges for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.5.3.2 Uniformity. Thirty rounds shall be assembled with the calculated charge weight of propellant. Ten rounds each shall be conditioned to -65°F, 70°F and 145°F (as indicated by special thermocoupled round). Test rounds fired at 70° shall be fired alternately with calibration rounds and shall be corrected

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to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.9.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results.

4.5.3.3 Ignition delay. Ignition delay, the time from primer initiation to a rise in pressure equivalent to ten percent of the maximum expected pressure, shall be measured by rear tourmaline gage for all the firings for information.

4.5.3.4 Pressure/temperature coefficient. The calculation of pressure/temperature coefficient shall be based on observed chamber pressure data obtained in the uniformity firings. The pressure/temperature coefficients between 70°F and 145°F and between 70°F and -65°F shall be calculated as follows:

$$PTH = \frac{A - B}{75^{\circ}F}$$

$$PTL = \frac{B - C}{135^{\circ}F}$$

Where PTH = Pressure/temperature coefficient between 70°F and 145°F, psi/°F

PTL = Pressure/temperature coefficient between 70°F and -65°F, psi/°F

A = Observed mean chamber pressure, psi at 145°F
B = Observed mean chamber pressure, psi at 70°F
C = Observed mean chamber pressure, psi at -65°F

4.5.3.5 Retest criteria. A lot shall be retested whenever any of the following conditions exist.

a. The final adjustment in charge weight exceeds one percent of the charge weight used in uniformity firings, retest for charge establishment in accordance with 4.5.3.1.

b. If for any reason, the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested.

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c. Test lot fails pressure requirements and copper crusher gage peak pressures are not consistent with velocity and piezo gage peak pressure data (for example: a high copper gage pressure is observed on a test round that shows an average velocity and normal piezo peak pressure).

4.5.3.6 Referral criteria. A test lot shall be referred to the procuring activity with a copy of test data forwarded to the technical agency for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. Calibration charges fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 25 feet per second of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration charges fired is less than specified.

e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70° and extreme value is an outlier (see 6.11).

5. PACKAGING

5.1 Packing.

5.1.1 Level A The propellant shall be packed in accordance with Dwg. 9390418-1.

5.1.2 Level B The propellant shall be packed in accordance with Dwg. 9390418-2.

5.1.3 Level C The propellant shall be packed in accordance with Dwg. 9390418-3.

5.2 Marking.

5.2.1 Levels A, B, and C Mark in accordance with drawing 9390418.

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6. NOTES

6.1 Intended use. The propellant covered by this specification is intended for use in the M203A1 propelling charge.

6.2 Aquisition requirements. (See MIL-A-48078)

6.2.1 Contractor's quality program. Contracts should identify MIL-Q-9858 as the applicable higher level contract quality control requirement and specify data item description DI-R-5297A.

6.2.2 Data requirements. One copy of the Propellant Description Sheet and one copy of all ballistic test data for each lot of propellant shall be forwarded to ARDEC, Picatinny Arsenal, NJ 07806-5000, ATTN: AMSMC-QAR-R(D), and SMCAR-AEE-BP. Individual measurements of the total volatiles, stick weights, stick bundle weights and dimensions shall be recorded in the Propellant Description Sheet for proper evaluation of stick propellant. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.3 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.4 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

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6.5 Proving ground test summary

TABLE V

Test Phase	Temperature	Sample size		Velocity Requirements		Press Req'ts*
				Service Velocity	Lot Std Dev of vel	Perm Ind Max Press
	(°F)	Cal	Test	(fps)	(fps)	(psi)
Charge Establishment	70	3	9	2710	N/A	57,950
Uniformity	-65		10	N/A	N/A	57,950
	70	10	10	2710	7.0	57,950
	145		10	N/A	N/A	57,950

* See 4.4.2.5.

N/A Not applicable

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6.6 Test anomalies. Any unusual ocurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.7 Permissible individual maximum pressure (PIMP). The weapon PIMP of 58,800 psi is applicable only when firing with an M549 projectile having the maximum weight. When an M549 projectile weighing 96.0 pounds is used, the applicable PIMP is calculated to be 57,950 psi.

6.8 Temperature indicator strips. The highest temperature reading from each indicator strip shall be recorded for informational purposes and reported on the applicable firing record.

6.9 Charge manufacture. All test and calibration charges shall be manufactured in accordance with MIL-C-70462.

6.9.1 Test charge. The test charges shall be loaded by the proving ground and the proving ground will furnish a complete set of operating procedures (including inspections to be performed and gage drawings) for approval to the technical agency prior to any manufacturing of test charges.

6.9.2 Calibration lot. All calibration lots shall meet with lot formation criteria of para. 4.4.1 of MIL-C-70462 except no mc than one lot of CBI or BP shall be used. Because of the non-availability of long term storage containers (metal) for stick propellant and combustible case components, and in order to maintain homogeneity, the calibration lot shall be loaded by the contractor responsible for LAP of this charge in production. The entire lot will be loaded and sealed in PA 103 containers at the same time (no further proving ground testing is required to accept the calibration lot). However, charges needed for establishment testing of calibration lot may be loaded at the proving ground.

6.10 Retest. All original and retest data shall be submitted to ARDEC, ATTN: AMSMC-QAR-R (D), Dover, NJ 07806-5000.

6.11 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_c) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

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Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10}= test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_c.

6.12 Reference stick propellant for closed bomb test. The contractor shall establish a standard stick propellant lot, normally, the calibration lot for reference for closed bomb test on stick propellant. The standard stick propellant lot shall meet the same requirement as specified by this specification and shall be approved for use as a reference by Contracting Officer Representative, Quality Assurance.

6.13 Blending process control document. The contractor will submit the blending process control document to the contracting officer with a copy of the submission forwarded to ARDEC, Dover, NJ 07806-5000, ATTN: AMSMC-QAR-R (D), and SMCAR-AEE-BP.

6.14 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and indentified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.16 Subject term (key word) listing.

Stick propellant
Propelling charge
Howitzer
Projectile

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1376-A352)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70456C (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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